

The Utility Problem

The ongoing end-consumption share displacement of carbon-heavy energy sources (like gas, wood or coal) by electrical power brings constraints on the capacity of the existing power delivery/distribution infrastructure. From the perspective of the medium voltage (MV) grid, not only the number of these points of consumption is increasing, but also the power required by them. The previously required demands in single-digit megawatts (MW) are becoming demands up to tens of MW, which are required by individual customers. This leads to previously unimaginable situations in the MV grid - causing the drastic reduction in available power carrying capacity of existing conductor lines, transformers and overall HV/MV substations.

A common solution to increased demands is to reinforce the grid or build new infrastructure. This process is, however, very costly and takes a long time. A faster and cheaper strategy to prevent the grid infrastructure from reaching the capacity limit is to optimize the grid appropriately. One option, which is commonly used in European types of grids, is to reconnect medium-voltage/low-voltage (MV/LV) stations to high-voltage/medium-voltage (HV/MV) stations where this is possible with the intention to handover or split the demand. For example, these HV/MV stations may have exhausted their capacity, so by connecting the MV/LV station to another HV/MV station, the demand will be handed over, and it will not be necessary to replace the transformer in this capacity-reached station with a larger one or to build a new HV/MV station. This will save significant investment costs.

However, this problem cannot be solved properly without appropriate data. It is also necessary to automate all the processes because it is impossible to go through thousands of stations manually to find suitable stations for optimization (reconnection) and possible grid modifications.

The Data Solutions and Methods

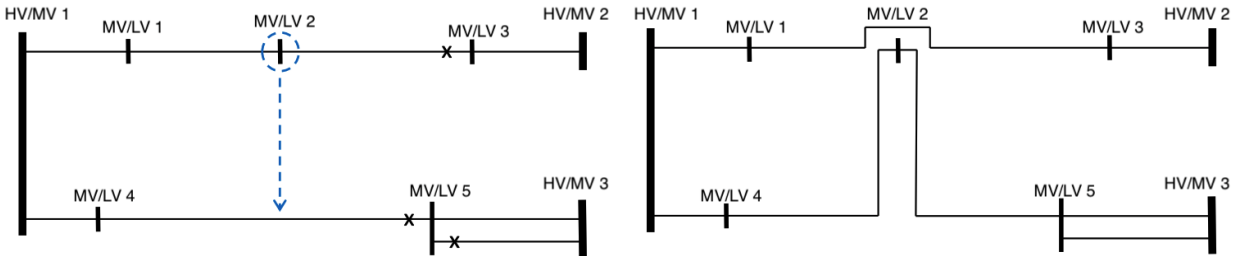
Above mentioned optimization is not simple, and possible reconnection of the conductor lines or substations can only be applied in some cases. Two such examples are used in this use case.

1. "Moving" an MV/LV station to another MV power line:

The first considered action is to “move” one station from one line to another line. The position of the moved station naturally (geographically) remains the same, but it is connected to another line - see figure below. The load on the originating line will therefore be lower, while the load on the receiving line will increase.

MV Grid Optimization Analysis

[UC33-01] - Awesense Energy Transition Platform Use Case



Current status and plan

New status

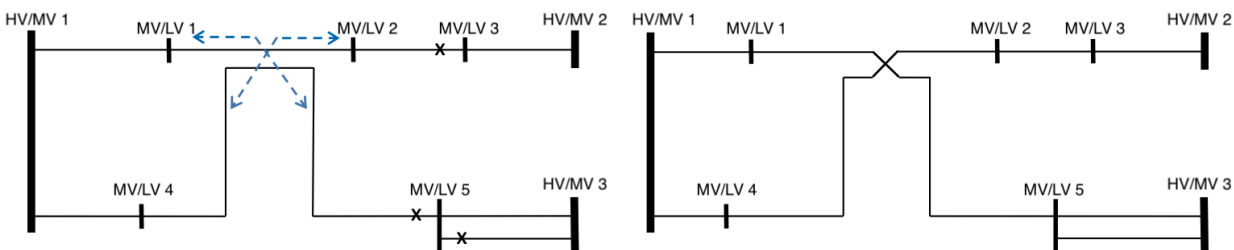
Legend: HV/MV - high/medium voltage station
MV/LV - medium/low voltage station

The general analysis procedure consists of the following steps:

- determining the shortest distances between the station and topological line segments of which the station is not a part,
- selecting distances shorter than X km(or miles) (the distance is measured as the crow flies, i.e. without taking into account buildings, other infrastructure and routes of the existing network, and thus represents the shortest theoretical distance),
- checking the allowable load on the lines (represented by the current situation modified by the relocation of the relevant station).

2. Reconnection in concurrence of conductor lines between existing feeders:

The second case is a reconnection made in the concurrence of two sections that now belong to two different feeders and run at least partly on a common route - see figure below. This will create two new feeders with different load sharing - lower loads on one part (i.e. on some sections) and higher loads on others.



Current status and plan

New status

The general procedure for finding a possible reconnection with conductor lines running concurrent consists of the following steps:

- defining (determining the geographic coordinates) of the considered reconnections in the cable/conductor concurrency,
- identification of the reconnection options for each considered concurrent reconnection,
- identification of the new feeders for each reconnection option and evaluation of the acceptability of each line in terms of feeding the different HV/MV stations,
- checking the allowable load for the new connection variants.

The unique integration of geospatial data (GIS) with time series data (data from stations) in the Awesense Platform enables this type of analysis. With the Platform, all the data you need is in one place and immediately available. Data is always up-to-date, error-free and synchronized.

Pre-built tracing functions help with the creation of this use case. Thanks to them, you are able to identify all stations on the MV feeder very quickly. Thanks to the integration of time data, you can then easily get the load of the selected stations and consequently evaluate the capacity limits/constraints. If the measurement is unavailable at a station (missing sensor or sensor failure), it is possible to use load estimation using the consumption of the meters belonging to that station (this is described in detail in another use case). The geospatial nature of the stored data allows for geospatial operations like nearby distance or various spatial joins. With these techniques, the development of such a use case is easy and fast.

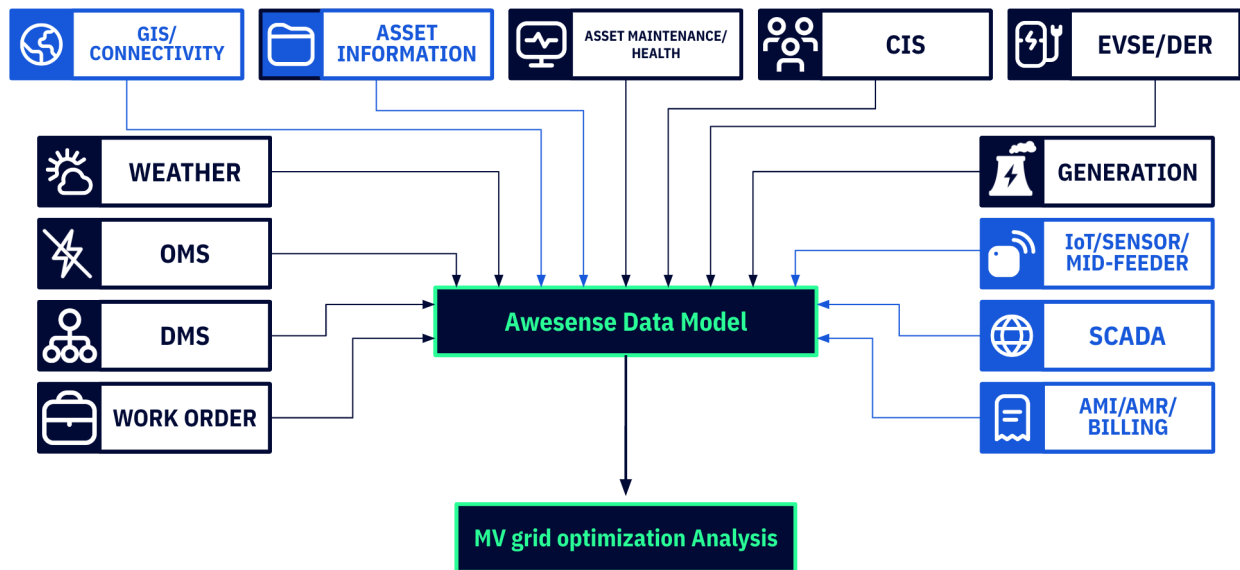
MV Grid Optimization Analysis

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Required Data Sets

The required data for this use case are visualized in the following diagram. Data sources highlighted in [royal blue](#) are required for this use case.



Available Use Case Implementations:

Users of the Awesense Energy Transition Platform can access this use case by the following means:

1. GitHub link to the Jupyter notebook can be provided upon request.